IS THE PLANET GETTING GREENER?

BBC MORE OR LESS | PODCAST SUMMARY

4.6%	20%	23%
GLOBAL LEAF AREA	**CHINA'S LEAF AREA**	**ARID + SEMI-ARID**
Approx. increase over 20 years	Approx. increase over 20 years	Share of global greening

HOW IS GREENING MEASURED?

Satellites analyse light reflected and absorbed by plants. Scientists use the **Leaf Area Index (LAI)** to estimate leaf area relative to ground area.

WHY IS IT HAPPENING?

CO2 fertilisation • Tree planting • Longer growing seasons in some colder regions • Agricultural changes

WHAT ABOUT DRY REGIONS?

Most greening occurs in wetter regions. The 23% figure means arid and semi-arid regions account for a share—not the majority—of global greening.

WHY IS MORE CO2 NOT ALL GOOD?

Climate change raises risks from heatwaves, droughts and sea-level rise, harms ecosystems and can reduce crop yields. Plant growth cannot cancel human-caused emissions.

TAKEAWAY: Earth is greening, but the global increase is about 4.6%—not 20%.
A greener planet is not necessarily a healthier planet.



Are Women 14 Times More Likely to Die in Natural Disasters?

Let's look at the real data.



Data and Statistical Methods

Compare mortality across countries.

What the Research Says

Results vary by country & disaster.

Real Examples of Natural Disasters

Bangladesh Cyclone (1991)	Haiti Earthquake (2010)	Hurricane Katrina (2005)
= 4.7×	= 0.9×	= 0.9×

Limitations of the Research

- No strong uniform evidence.
- Results differ by region.

Real-Life Applications

Helps improve preparedness & policy.

Result:

Death rates vary — not **14×**.

Takeaway:

A claim cannot be established by just observation, but through **proper evidence**, but even in such cases we should account for **other factors**.

- Not just women — sometimes children too.
- Men are higher risk in developed countries; women in developing countries.

Source: BBC, "Are women 14 times more likely to die in natural disasters?", Oct. 30, 2018.

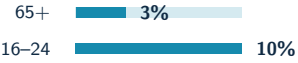
Is There Really a Loneliness Epidemic?

THE QUESTION Young people report loneliness more often than older people. Does that prove loneliness is increasing over time?

01 AGE SNAPSHOT

England · ONS Community Life Survey, 2016–17

Often / always feel lonely



Five response options: often/always, sometimes, occasionally, hardly ever, never.

02 CHANGE OVER TIME

US high-school students · 1977–2012

Mean score from six loneliness items

Scale: 1 = disagree, 5 = agree



Schematic trend only; the source chart reports a very small decline. Its vertical axis is truncated.

03 THREE WAYS TO ANALYSE

Cross-sectional

Compare age groups at one time. Describes differences, not trends.

Longitudinal

Follow the same people as they age.

Cohort comparison

Compare gen at similar ages.

Source:https://ourworldindata.org/loneliness-epidemic

Should You Follow the 5-Second Rule?

G 30

Does faster pickup prevent bacterial contamination?

Methods & Analysis

Statistical Analysis (t-Test)

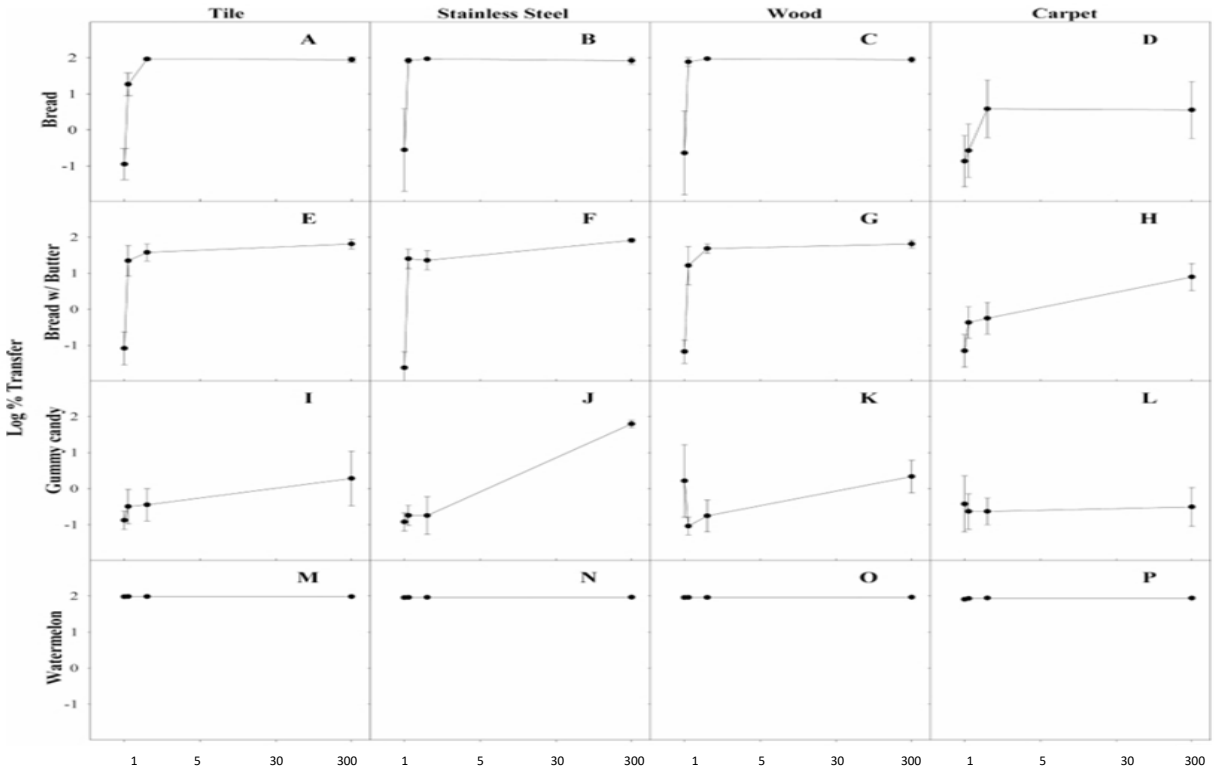
Computed sample mean ($\bar{X}$) and sample variance (S^2) for each food group. Used two-sample t-tests to evaluate whether differences across contact times are statistically significant.

Method & Sampling

Tested 4 food types (watermelon, bread, bread with butter and gummy candy) dropped on contaminated surfaces across 4 contact durations(0s control, 5s, 30s, 300s) with measured bacterial counts (CFU/g).

Primary Variables / factors

Contamination levels depend heavily on four factors: contact time, food type, surface type, and the bacterial preparation method.



Limitations

Conducted in controlled laboratory conditions with specific food items and bacteria. Lab counts do not directly equate to infectious dose or actual illness risk in home environments.

Key Results

Bacterial contamination increases with contact time (Watermelon> bread with butter> bread > Gummy candy). Bacterial transfer occurs instantaneously (<1s),disproving the absolute 5-second safety rule.

Takeaway

The 5-second rule is not a simple yes/no threshold. Picking up food faster reduces relative contamination, but surface cleanliness and food moisture dictate real safety.

Source: BBC Radio 4, More or Less, "Should You Follow the 5-Second Rule?" (2022).
Audio timestamps: Intro 0:30-2:45 | Key finding / Data 22:50-28:15

Miranda, R. C., & Schaffner, D. W. (2016). Longer contact times increase cross-contamination of Enterobacter aerogenes from surfaces to food. https://doi.org/10.1128/AEM.01838-16

The Medical Test Paradox & Bayesian Inference

UDIT SRIVASTAVA
ADITYA MAHANTA

1. DEFINITIONS & BASELINE COHORT

Prevalence / Prior P(D): Base rate of disease in population = **1%** (10 in 1,000).

Sensitivity P(+|D): True positive rate; test accuracy given disease = **90%**.

Specificity P(-|~D): True negative rate; accuracy given healthy = **91%**.

False Positive Rate P(+|~D): Healthy testing positive = **9%** (100% - Specificity).

PPV P(D|+): Positive Predictive Value; actual chance of disease given positive result.

1,000-Woman Cohort Frequency Breakdown:

Test Result	Cancer (10)	Healthy (990)	Total
Positive (+)	**9 (True +)**	**89 (False +)**	**98**
Negative (-)	1 (False -)	901 (True -)	902

Core Problem: Because disease is rare (1%), the total false positives (89) far outweigh true positives (9).

2. MATHEMATICAL CALCULATIONS

METHOD A: STANDARD BAYES' RULE (PROBABILITIES)

$$P(D|+) = [\ P(+|D) \times P(D)\ ] / P(+)$$

- Numerator = 0.90 × 0.01 = **0.009**
- Denominator = (0.90×0.01) + (0.09×0.99) = **0.0981**
- P(D|+) = 0.009 / 0.0981 = **9.17%**

(Frequency table gives 9 / 98 = **9.18%** due to rounding 89.1)

METHOD B: REDESIGNED ODDS FORM

Posterior Odds = Prior Odds × Bayes Factor

- **Prior Odds:** Odds of disease before test - [**P(D)/P(-D)**].
- **Bayes Factor:** How much a positive result shifts the odds (**sensitivity/ false-positive rate**).
- **Posterior Odds:** Odds of disease after a positive result.

- **Prior Odds:** 1% prob = **1 : 99**
- **Bayes Factor:** Sens / FPR = 90% / 9% = **10**
- **Posterior Odds:** (1 : 99) × 10 = **10 : 99**
- **Posterior Prob:** 10 / (10 + 99) = **9.17%**

Gigerenzer Seminar Finding: >50% of practicing gynecologists incorrectly answered 90%, confusing Sensitivity P(+|D) with PPV P(D|+).

3. CONTEXT SHIFT & APPLICATIONS

Impact of Base-Rate Prevalence Shift:

- **At 1% Prevalence:** Prior 1:99 × 10 = 10:99 Posterior = **9.17% PPV**
- **At 10% Prevalence:** Prior 1:9 × 10 = 10:9 Posterior = **52.6% PPV**
- ***Insight:*** *Diagnostic value changes radically based on patient population.*

Critical Evaluation & Limitations:

- **Hypothetical Model:** Demonstrates mathematical principles rather than direct real-world patient outcomes.
- **Dynamic Parameters:** Prior prevalence and sensitivity vary by age, symptoms, and clinical setting.

Cross-Domain Applications:

- **Fraud Detection:** Rare fraud leads to high false alarms despite high classifier accuracy.
- **Airport Security:** Rare threats mean alerts require contextual verification.

Takeaway: Always report PPV/NPV alongside target population prevalence.

Key Takeaway: A positive test is evidence that updates prior belief, not a final diagnosis.
To interpret evidence correctly, combine test accuracy (Bayes factor) with prior risk (base rate).

Posterior Odds = Prior Odds × Bayes Factor

Does the news reflect what we die from?

Actual causes of death vs. media coverage in the US (2023)

ACTUAL DEATHS

56%

Heart disease (29%) + Cancer (27%)

NYT MEDIA COVERAGE

7%

Heart disease (29%) + Cancer (27%)

Where does media attention go?

Share of media coverage devoted to each cause

News outlet	HOMICIDE	TERRORISM
New York Times	42%	18%
Washington Post	46%	12%
Fox News	52%	11%

Overrepresentation in NYT coverage

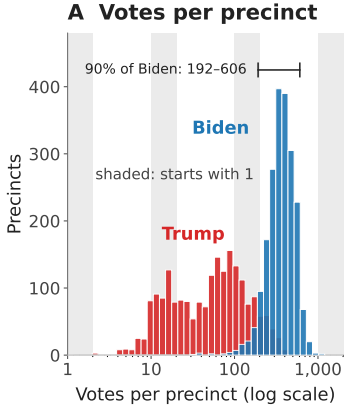
43× HOMICIDE

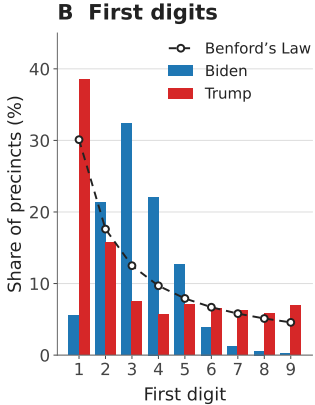
18,240× TERRORISM

15 selected causes account for approximately 76% of US deaths in 2023.
Source: US CDC, Media Cloud, Global Terrorism Index; Ritchie et al. (2025), Our World in Data.

Do Chicago's 2020 votes really break Benford's Law?

In 2020, people claimed Biden's Chicago votes were fake because they don't follow Benford's Law.





Methods

First-digit and last-digit frequency checks on Chicago precinct vote counts.

Key result

Precincts are similar in size, so Biden's first digits can't follow Benford.

Takeaway

Benford's Law can flag odd data, but it can't prove fraud.

Media: M. Parker, *Why do Biden's votes not follow Benford's Law?*, Stand-up Maths, YouTube, 10 Nov 2020 (youtu.be/etx0k1nLn78)
Timestamps: 0:01–3:40 Benford's Law · 3:41–6:26 forensic accounting · 6:27–10:49 Chicago election data · 10:50–15:46 last-digit check
Figure: re-plotted in Python from Chicago Board of Election Commissioners 2020 precinct results (2,069 precincts)

EXPECTED GOALS & STATS IN FOOTBALL | THE TRENDS

A scoreline tells us what happened - xG helps explain the quality of chances created.

What xG means?

Expected Goal (xG): Probability that a shot will become a goal.
Range: 0 (Lower) ----------1 (Higher)
xG of 0.7 = 7 goals out of 10 similar shots

How is xG calculated?

- Shot distance & angle
- Velocity & Trajectory
- Shot type
- Shot position wrt the ground
- Defenders, attackers & keeper position

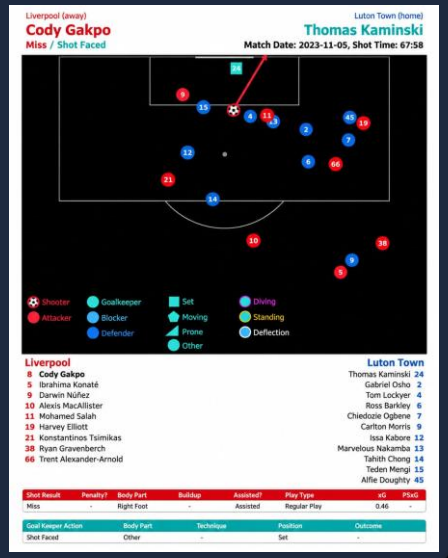


Limitations

- xG measures chances, not certainty
- Model definitions differ
- Some actions are unmeasured.

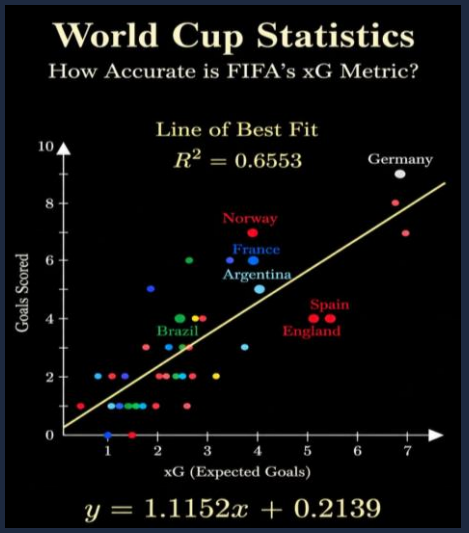
Same Area ≠ Same Chance



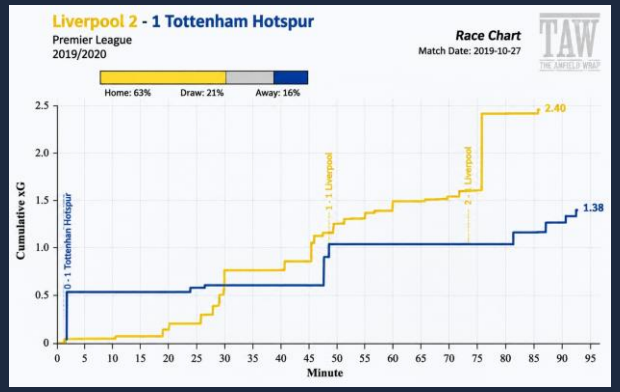


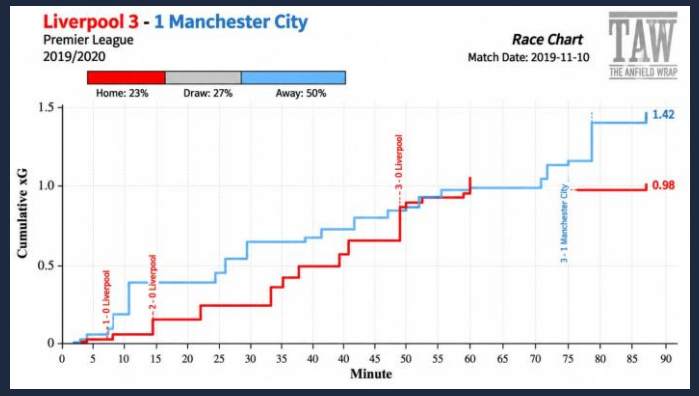
Source: Statsbomb

The Trends



Race Chart





Source: The Anfield Wrap

xG → Scorelines → Match Context → Longer-term Performance Trends

From One Shot To A Season-long Trend

- Over many matches, xG helps reveal performance trends that the scoreline can hide.
- Predicting future team performance.
- Across matches, xG trends can reveal more than one scoreline.

Key Takeaway

- Use xG with match context and observation - not as a verdict.

Media: The Anfield Wrap (TAW) "Expected Goals & Stats In Football, The Trends" | YouTube & Spotify | 24 July 2024 | ~24min

By: Eric P. Karr [CE25M827] & Gemechis Teferi Benti [CE25M812]

Can Random Selection Explain the Absence of Conservative MPs at PMQs?

Could none of 47 Conservatives appear among 15 names by chance?

SHUFFLE DATA | **311** applicants | **47** Conservative | **264** Non Conservative | **15** randomly selected

TWO METHODS, ONE PROBABILITY

Hypergeometric distribution

Count favourable selections / all possible selections

$$P(X = 0) = C(264, 15) / C(311, 15)$$

Sequential conditional probability

Multiply the changing probabilities without replacement

$$P(X = 0) = (264/311) \times (263/310) \times \ldots \times (250/297)$$

SAME RESULT **8.05%** ≈ 1 in 12 comparable draws

METHODS

Hypergeometric counting and conditional probabilities yield identical results.

KEY RESULT

Zero Conservatives among 15 selections has an 8.05% chance.

TAKEAWAY

Same model audits admissions lotteries, conditional on who applies.

Source: More or Less, “Does Europe suffer more heat deaths than the US?”, BBC Radio 4, 17 Jun 2026; PMQs segment 14:46–19:01, calculation 16:22–18:28. Shuffle procedure: UK Parliament.

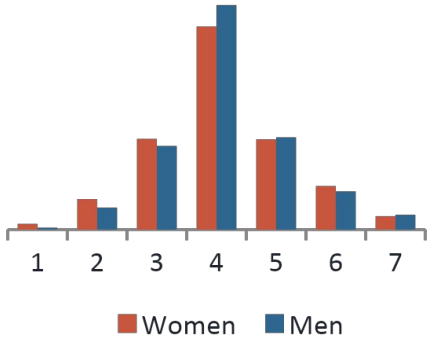
Is “Women’s Winter” Real? Testing a Podcast Claim with Data

Ayush Sourav Jagaty (MA26M007) • Roki Howladar (MA26M016) • MA 5104 Statistical Inference and Learning, IIT Madras

Do women really vote colder in offices, or do they just sit in colder rooms?

Welch t-test: vote by gender

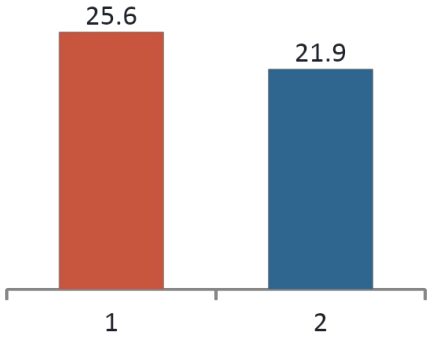
H_0: mean vote, women = men
H_1: mean vote, women ≠ men



Mean vote 0.101 vs 0.165 (t = −5.62, p < 0.001): reject H_0,

Chi-squared: feels cold

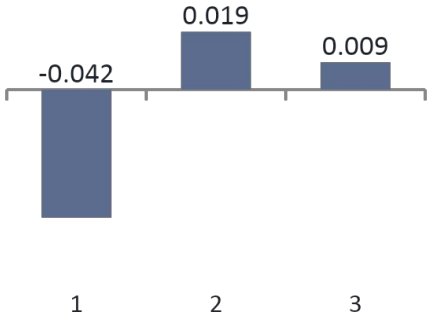
H_0: cold is independent of gender
H_1: cold depends on gender



$\chi^2 = 72.4$, $p < 0.001$: reject H_0.

Regression: female effect

H_0: β(female) = 0, adjusted
H_1: β(female) ≠ 0, adjusted



Women’s rooms ran 0.55°C cooler. Adjusted, p = 0.42: cannot reject H_0

Methods

Welch t-test, chi-squared test, and regression on ASHRAE office data (n=38,403).

Key result

Women feel colder (25.6% vs 21.9%); adjusting for room temperature erases it.

Takeaway

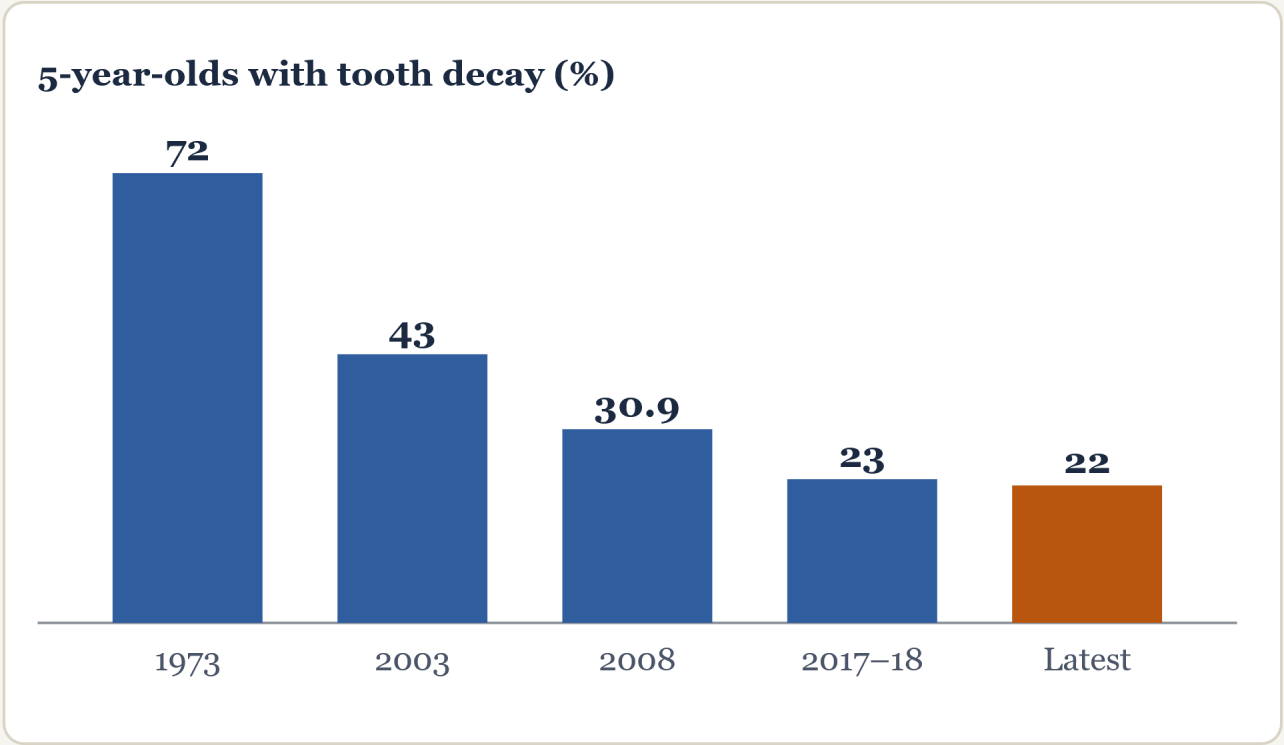
Real but small, mostly cooler rooms; not true for every woman.

Limits: *observational pooled surveys; metabolic rate assigned, not measured; podcast lab study had only 16 women .*
Fix: compare within the same building.

Media: BBC Radio 4, More or Less, “Saturday at 15:08” Data: ASHRAE Global Thermal Comfort Database II, office rows with recorded gender (n = 38,403).

Bad teeth and A&E: separating hospital admissions from A&E attendances

An MP's claim mixed up two different counts (02:00–02:40).



Methods: Compare admission rates per 100,000 with school dental survey percentages

Key result: Claim confused admissions with A&E; decay tops admissions but is falling

Limits: Little A&E data on dental causes; old surveys not directly comparable

Use elsewhere: Any "number one cause" claim: road injuries, school absence

Takeaway: Check what a headline counts before trusting the number

Source: More or Less, BBC, 16 Sep 2026 · Timestamps: 00:40–01:00 · 02:00–02:40 · 04:00–04:40 · 05:40–07:20

GDP and India's Economic Well-being

A Statistical Analysis

Jitendra Kumar (MA26M010) & Aditya Kumar (MA26M001)

1. Descriptive Statistics

$n = 26$ (Number of observations)

$\bar{g} = 6.25\%$ (Mean growth) — Med $= 7.31\%$ (Median)

$s = 2.98$ pp (Standard deviation)

Min: -5.78% (2020) — Max: 9.69% (2021)

2. Hypothesis Testing

Compare mean GDP growth across periods:

$$H_0 : \mu_1 = \mu_2 \qquad H_1 : \mu_1 \neq \mu_2$$

Statistic	Value
Mean (2000–2010)	6.49%
Mean (2011–2020)	5.21%
Welch's t	0.888
p-value	0.391

Decision: Fail to reject H_0 at $\alpha = 0.05$.

3. Linear Regression

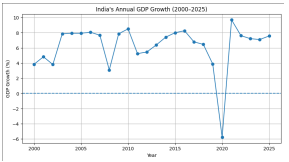
$$g_t = \beta_0 + \beta_1 t + \epsilon_t$$

Slope: 0.00345 percentage points/year — $R^2 = 0.000078$, $p = 0.966$

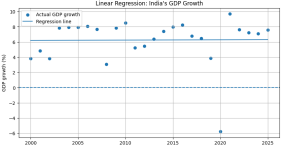
4. Preliminary Findings

- **Period Means:** No statistically significant difference in mean GDP growth across periods ($p = 0.391$).
- **Linear Trend:** No significant long-term upward or downward trend over time ($p = 0.966$).

5. GDP Growth Over Time



6. GDP and Well-being



Source: World Bank, World Development Indicators (WDI). Annual observations may be temporally dependent.

Why Cricket Defeats Hurt Indian Markets More Than Wins Help

India won the 2026 T20 World Cup final, yet markets fell nearly 2% on Monday.

Average next-day Nifty fall, by match result



Loss effect: over 7× the win effect (06:40–06:52)

SAME MONDAY, OTHER SHOCKS

Middle East escalation (01:42), Korea –10% (02:06), oil $115 (02:12), rupee past 92 (02:18)

METHODS

Next-day market returns compared after wins, losses and rivalries.

KEY RESULT

Losses move Indian markets over seven times more than wins.

TAKEAWAY

Sentiment moves markets, but confounders must be ruled out first.

Source: “Raghav’s Take”, The Quint, March 2026. Chart redrawn from on-screen Hindustan Times figures (win “flat”, loss 0.231%).
Timestamps: 02:22 · 04:02–04:38 · 05:00–06:06 · 06:40–06:52 · 07:08–08:08 · 09:02–10:16